Web-Based Malayalam Keyboard Testing Report

Date: October 26, 2023

Test Url: https://gistlangserver.in/smartkeyboard/demonstration

Objective: To ensure the web-based Malayalam keyboard developed by C-DAC is accurately translates keystrokes into correct Malayalam script.

Detailed Findings: please find below details of the issues observed during functionality testing the keyboard.

	Observation	**Description**
01	Unicode coverage	The keyboard is missing some of the Malayalam Unicode. Eg (0D3D, 0D4C, 0D4E, 0D4F, 0D54-OD5F, OD60-OD63, OD70-OD78)
02	Chilly generation issues	The automatic conversion of consonant+virama to a chillu letter was applied incorrectly in some cases. which 0D30 + 0D4D => CHILLU RR (0D7C) 0D28 + 0D4D => CHILLU N (0D7B) 0D23 + 0D4D => CHILLU NN (0D7A) 0D32 + 0D4D => CHILLU L (0D7D) 0D33 + 0D4D => CHILLU LL (0D7E)
		consonant+virama+consonant is not properly coming as the auto chillu rendering is happening. Eg. 1. 0D32 (ല) + 0D4D (്) +0D32 (ല) is rendering as 0D7D (ൽ) + 0D32 (ല) 2. 0D2A (ക) + 0D4D (്) +0D32 (ല) + 0D47 (േ) is rendered as 0D2A (ക) + 0D4D (്) + 0D47 (േ) +0D32 (ല)
03	Frequent used words	The vowel sign 0D43 (ൃ), 0D47 (േ) should be added in the autogenerated consonant list in top. Instead of ് 0D01, ഃ 0D03 as these are less frequently used .
04	Incorrect Glyph Spacing	overlapping with the adjacent characters and Visible white space between a consonant and its vowel sign eg. 1. 0D4D + 0D30 (ർ) is show with white spacing 2. 0D4D + 0D35 (വ) is overlapping with previous characters 3. 0D4D + 0D0B (ഋ) is also overlapping with previous characters
05	Incorrect Rendering of Reph	The sequence Consonant + Virama + RA (e.g., 0D2A + 0D4D + 0D30) should form a "Reph" which is rendered before the base consonant. This is same with all the consonants
06	Failure to form conjuncts with La (ല) Virama	Rendering failed on CONSONANT + VIRAMA + LA (0D4D + 0D32) correctly. Eg. 0D17 (ഗ) + 0D4D (്) + 0D32(ല) + 0D3E (ാ) + 0D38 (സ) + 0D4D (്)

07	Failure to Render Common Conjunct "ന്വ"	The text rendering engine fails in the sequence ന + VIRAMA + വ (0D2E + 0D4D + 0D2A). This is one of the most frequently used conjuncts in the Malayalam language.

Other recommendations:

1. Include support for the Unicode control characters Zero-Width Joiner (ZWJ - U+200D) and Zero-Width Non-Joiner (ZWNJ - U+200C) into the keyboard for advanced users.
2. Also the language Toggle Button does not change its label dynamically between EN & ML

References

The Unicode Standard, Version 16.0 (https://www.unicode.org/charts/PDF/U0D00.pdf)
Malayalam Standardization Report (https://keralabhashainstitute.org/downloads/msr.pdf)

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